

Le Nghia Hiep

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SUMMARY

A Computer Engineering student with a sharp systems-oriented mindset and a passion for core technology research. I possess a strong technical foundation, which enables me to quickly adapt to and master new technologies to solve complex problems. Beyond technical expertise, I take pride in my strong communication, presentation, and teamwork skills.

EDUCATION

Hanoi University of Science and Technology (HUST)

Majoring in Computer Engineering - CPA 3.60

Hanoi, Vietnam

Sep. 2023 – Present

Phu Xuyen A High School

Achieved 28.7 scores (Group A00) in the High School Graduation Examination

Phu Xuyen, Hanoi, Vietnam

2020 – 2023

EXPERIENCE

Modern Computer Networks and Communications Technology Lab - NetLab

Research Assistant

SoICT - HUST

April, 2025 – Present

- Research on AIoT in Healthcare
- Biosignal Research (EEG, ECG)

TECHNICAL SKILLS

Language: C/C++ (Embedded System), Python (Data processing & AI), Java, JS/HTML/CSS

Embedded Systems & Hardware: ESP32 development, Sensors integration

Networking & Protocols: IoT Protocols (MQTT, HTTP/Restful API), Network Basics (TCP/IP, UDP)

Tools & Platforms: Git/Github, Arduino IDE/PlatformIO

English: Intermediate (TOEIC Equivalent 650-700, enough to read and understand AI papers and documentation)

PROJECTS

SmartClock - AI-Powered IoT Health & Environment Monitor

ESP32, PlatformIO, FreeRTOS, Python, FastAPI, MQTT, Scikit-Learn, SQLite, Kotlin

Full-stack AIoT project

- Designed and architected an end-to-end IoT ecosystem connecting edge hardware, an AI-powered cloud backend, and a native mobile application.
- Developed non-blocking firmware for the ESP32-C3 microcontroller using asynchronous timing loops, ensuring high-frequency sensor polling (MAX30102, SHT31) and smooth 33 FPS OLED rendering without thread-blocking.
- Implemented a hybrid communication pipeline integrating real-time time-series data streaming via MQTT with secure user profile synchronization via RESTful APIs (FastAPI).
- Trained and deployed a Machine Learning model (Scikit-Learn) on a local Python server to process raw infrared (IR) PPG signals and return real-time health and recovery assessments.
- Engineered a responsive Android application using Kotlin and Jetpack Compose under the MVVM pattern, incorporating live data binding and dynamic canvas-based PPG waveform animations synchronized with real-time BPM.
- Optimized hardware power consumption by programming ESP32 deep-sleep modes, utilizing a capacitive touch button as an external GPIO hardware wake-up source.
- Device images: *Direct link to circuit image*
- Github: *link*

AI-Driven ECG Arrhythmia Classifier

ESP32, MQTT, Deep Learning, TensorFlow, Scikit-learn

Full-stack AIoT project

- Hardware Design & Integration: Designed schematics and developed prototypes integrating ESP32 microcontrollers, AD8232, and MPU-6050 sensors; optimized power management systems using TP4056 charging circuits and LiPo batteries.

- Communication Protocols & Connectivity: Implemented real-time data streaming using the MQTT protocol over wireless networks to ensure stable, low-latency communication between devices and the cloud.
- Hybrid Deep Learning Architecture: Designed and implemented a hybrid 1D CNN-LSTM model to classify cardiac arrhythmias. Used 1D CNN for spatial feature extraction from raw ECG signals and LSTM to capture long-term temporal dependencies in heart rate variability
- Signal Processing & Optimization: Developed a preprocessing pipeline to handle raw signal noise (Baseline Wander, Powerline Interference) using digital filters. Conducted data segmentation and normalization on the MIT-BIH Arrhythmia Database to ensure high model generalization
- Device images: *Direct link to circuit image*
- Source code: *CNN-1D-and-LSTM-for-Arrhythmia-Classification*

RAG-based University Query Chatbot

RAG, Python, LangChain, LLMs, FAISS, Streamlit

RAG Chatbot

- Data Pipeline Development: Engineered a robust pipeline to extract and preprocess data (including advanced chunking) from complex academic regulation PDFs
- Retrieval Optimization: Leveraged Recursive Character Text Splitting and Embedding Models to transform unstructured text into high-dimensional vectors, managed within a FAISS vector database
- Advanced RAG Implementation: Integrated Context Retrieval to provide automated source citations for every response, ensuring transparency and factual accuracy for legal and regulatory information
- Deployment: Developed and launched an interactive user interface using Streamlit to facilitate seamless student-bot interaction
- Source code: *AI-chatbot-for-HUST-students*

HONORS AND AWARDS

- *Top 10 — Kawai Business Startup 2026*
- *Second Prize in the Student Creative Ideas Challenge - SoICT 2025*
- *Awarded Academic Achievement scholarship of SoICT*
- *Awarded Lotte Scholarship, SCG Sharing the Dream Scholarship 2023, 2024*